

# REGULARISATION IN GUIDING

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## 1. GENERAL SETUP

Consider discrete-time nonlinear Gaussian model

$$X_{t+1} \mid X_t = x \sim \mathcal{N}(b_t(x), Q_t(x)).$$

Assume to observe

$$V_t \mid X_t = x \sim K(x, \cdot),$$

Consider the continuous-time SDE model

$$dX_t = b(t, X_t) dt + q dW_t, \quad X_0 = x_0.$$

Assume to observe

$$V_i \mid X_{t_i} = x \sim K(x, \cdot).$$

We assume the Markov kernel  $K$  to admit a density  $k(x, \cdot)$  with respect to Lebesgue measure.

Outline:

- (1) focus on filtering using SMC
- (2) Upon linear approximation of  $b$  and  $k$ , we can compute guiding functions
- (3) Analyse behaviour of resulting guided process, show need of safeguarding using  $\varepsilon$ .

## 2. DISCRETE EXAMPLE SETUP

We consider the Markov kernel  $P(x_0, dx_1)$  defined by

$$X_1 \mid X_0 \sim \mathcal{N}(\omega + \psi X_0, \eta^2).$$

We assume  $v \sim x + \log Z^2$ , where  $Z \sim \mathcal{N}(0, 1)$ . Define

$$h(x_1) = \text{density of } X_1 + \log Z^2 \text{ evaluated at } v, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

We have

$$P(X_1 + \log Z^2 \leq v) = P(Z^2 \leq e^{v-X_1})$$

so

$$h(x_1) = f_{\chi^2(1)}(e^{v-x_1}) e^{v-x_1}.$$

For large  $x_1$  it will behave like  $\sqrt{e^{v-x_1}}$ . It should also behave this way for large  $x_1$ . Define, using Gaussian approximation to the distribution of  $\log Z^2$

$$g(x_1) = \varphi(v; \mu + x_1, \sigma^2), \quad \mu = -1.27, \quad \sigma^2 = \pi^2/2,$$

where  $\varphi(\cdot; \mu, \sigma^2)$  denotes the Gaussian density.

We have twisted kernels

$$P^h(x_0, dx_1) = \frac{P(x_0, dx_1) h(x_1)}{\int P(x_0, dx_1) h(x_1)}, \quad P^g(x_0, dx_1) = \frac{P(x_0, dx_1) g(x_1)}{\int P(x_0, dx_1) g(x_1)}.$$

Define  $c_g = \int P(x_0, dx_1) g(x_1)$  and similarly  $c_h$ . As  $x_0$  is fixed in this section, we omit it from our notation. In our specific case

$$c_g = \varphi(v; \mu + \omega + \psi x_0; \sigma^2 + \eta^2).$$

Moreover,

$$P^g(x_0, dx_1) \sim \mathcal{N}(\mu^*, (\sigma^*)^2),$$

with

$$(\sigma^*)^2 = \frac{\eta^2 \sigma^2}{\eta^2 + \sigma^2} \quad \mu^* = \frac{\sigma^2(\omega + \psi x_0) + \eta^2(v - \mu)}{\eta^2 + \sigma^2}.$$

The measure obtained from  $g_\varepsilon$  is a mixture:

$$P^{g+\varepsilon}(x_0, dx_1) = \frac{g(x_1) + \varepsilon}{c_g + \varepsilon} P(x_0, dx_1) = \lambda_g(\varepsilon) P^g(x_0, dx_1) + (1 - \lambda_g(\varepsilon)) P(x_0, dx_1),$$

with  $\lambda_g(\varepsilon) = c_g / (c_g + \varepsilon)$ .

### 3. SEQUENTIAL MONTE CARLO APPROACH

Suppose we do sequential Monte Carlo to sample from  $X_1 \mid v$  and to find the marginal likelihood  $p(v)$ , which is actually  $c_h$ . We use mutation kernel  $M = P^{g+\varepsilon}$ . Then it follows from the results in Chapter 5 of Chopin-Papaspiliopoulos that the potential should be

$$\begin{aligned} G_\varepsilon(x_1) &:= h(x_1) \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} \\ &= h(x_1) \frac{\int P(x_0, dx_1) g(x_1) dx_1 + \varepsilon}{g(x_1) + \varepsilon} \\ &= h(x_1) \frac{c_g + \varepsilon}{g(x_1) + \varepsilon}. \end{aligned}$$

The estimator for the marginal likelihood is

$$\hat{L} := \frac{1}{N} \sum_{n=1}^N G_\varepsilon(X_1^{(n)}), \quad (X_1^{(n)})_{n=1}^N \sim P^{g+\varepsilon}(x_0, dx_1).$$

For any  $\varepsilon > 0$

$$\mathbb{E}_{P^{g+\varepsilon}} \hat{L} = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X_1) = \int h(x_1) \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} P^{g+\varepsilon}(x_0, dx_1) = c_h.$$

#### 3.1. Analysis of variance of $\hat{L}$ . Define

$$m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} [G_\varepsilon(x_0, X_1)]^2.$$

By expanding we get

$$\begin{aligned} m(\varepsilon) &= \int h(x_1)^2 \left( \frac{c_g + \varepsilon}{g(x_1) + \varepsilon} \right)^2 \frac{P(x_0, dx_1) (g(x_1) + \varepsilon)}{c_g + \varepsilon} \\ &= (c_g + \varepsilon) \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1). \end{aligned} \tag{1}$$

Define

$$I(\varepsilon) = \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1)$$

and

$$\bar{m}(\varepsilon) = \log m(\varepsilon) = \log(c_g + \varepsilon) + \log I(\varepsilon).$$

Note that we can evaluate  $c_g$  and  $I(\varepsilon)$  numerically using Gauss-Hermite quadrature.

Note that we can rewrite to

$$I(\varepsilon) = \int h(x_1) \frac{\int h(x_1) P(x_0, dx_1)}{g(x_1) + \varepsilon} P^h(x_0, dx_1) = c_h \int \frac{h(x_1)}{g(x_1) + \varepsilon} P^h(x_0, dx_1).$$

though we won't use this expression in the numerics.

**3.2. Infinite variance case.** Suppose  $\varepsilon = 0$ . Since  $c > 0$ , finiteness of  $\bar{m}(0)$  is equivalent to finiteness of

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1).$$

Using  $f_{\chi^2(1)}(u) = (2\pi u)^{-1/2} e^{-u/2}$  we get

$$h(x_1) = \frac{1}{\sqrt{2\pi}} e^{-(v-x_1)/2} e^{-e^{v-x_1}/2},$$

so

$$h(x_1)^2 = \frac{1}{2\pi} e^{-(v-x_1)} e^{-e^{v-x_1}}.$$

Writing  $t = x_1 - v$ , the integrand of  $I(0)$  equals

$$\frac{h(x_1)^2}{g(x_1)} p(x_1 | x_0) \propto \exp\left(t - e^{-t} + \frac{(t + \mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2}\right).$$

We analyse the tails:

- *Right tail*  $t \rightarrow +\infty$ . The term  $e^{-t} \rightarrow 0$ , and expanding the quadratics the dominant behaviour is

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right).$$

This is integrable if and only if the coefficient of  $t^2$  is strictly negative, i.e.

$$\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} < 0 \iff \eta^2 < \sigma^2.$$

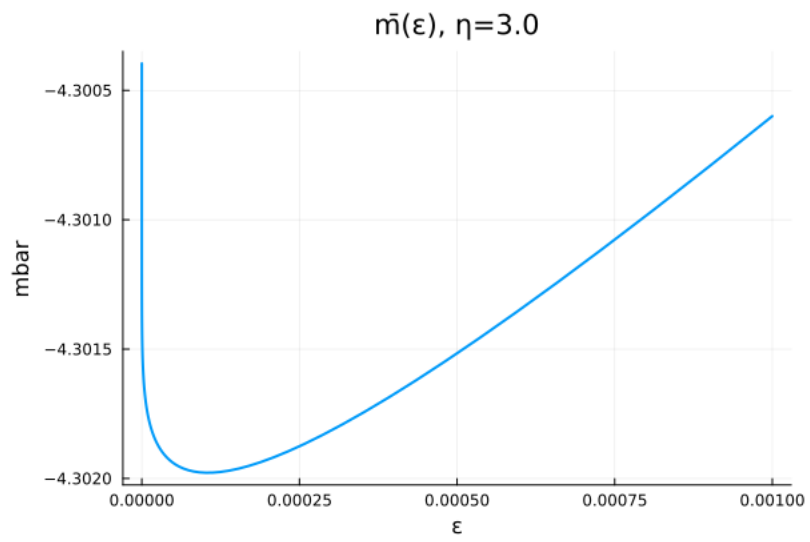
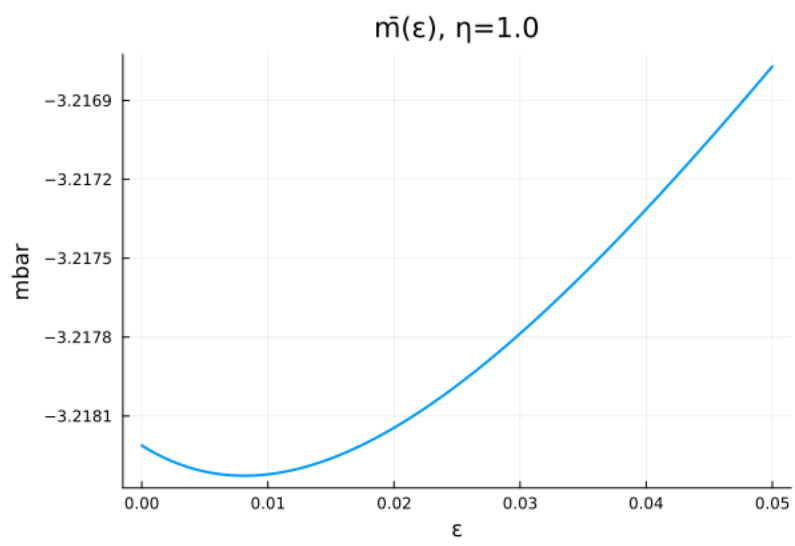
- *Left tail*  $t \rightarrow -\infty$ . The term  $e^{-t} \rightarrow +\infty$  dominates inside the exponent, sending the integrand to zero faster than any polynomial. Hence the left tail never causes divergence.

Therefore,

**Proposition 1.** With the observation scheme  $v = X_1 + \log Z^2$ ,  $Z \sim \mathcal{N}(0, 1)$ , we have

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1) < \infty \iff \eta < \sigma.$$

It follows that for  $\eta < \pi/\sqrt{2} \approx 2.221$  no regularisation is needed. For  $\eta = \sigma$  the integral diverges (though like  $e^t$ ), while for  $\eta > \sigma$  divergence is much faster.



#### 4. IMPORTANCE SAMPLING QUALITY: KL AND $\chi^2$ DIVERGENCES

**4.1. General theory.** The natural divergences for assessing IS quality of  $\hat{L}$  are the KL divergence and the  $\chi^2$  divergence between the target  $P^h$  and the proposal  $P^{g+\varepsilon}$ :

$$\begin{aligned}\text{KL}(P^h \| P^{g+\varepsilon}) &= \int \log \frac{dP^h}{dP^{g+\varepsilon}} dP^h, \\ \chi^2(P^h \| P^{g+\varepsilon}) &= \int \left( \frac{dP^h}{dP^{g+\varepsilon}} \right)^2 dP^{g+\varepsilon} - 1.\end{aligned}$$

Since

$$\frac{dP^h}{dP^{g+\varepsilon}}(x_1) = \frac{h(x_1)/c_h}{(g(x_1) + \varepsilon)/(c_g + \varepsilon)} = \frac{G_\varepsilon(x_1)}{c_h},$$

we have

$$\chi^2(P^h \| P^{g+\varepsilon}) = \frac{m(\varepsilon)}{c_h^2} - 1,$$

so finite variance of the IS estimator is *exactly equivalent* to  $\chi^2(P^h \| P^{g+\varepsilon}) < \infty$ . The  $\chi^2$  divergence thus directly controls the variance. Note, using  $\log x \leq x - 1$  (and thus  $2 \log x \leq x^2 - 1$ )

$$2\text{KL}(P^h \| P^{g+\varepsilon}) \leq \int \left( \frac{dP^h}{dP^{g+\varepsilon}} \right)^2 dP^h - 1.$$

Original bound I wrote down was wrong.

**4.2. KL divergence: explicit form.** We have

$$\text{KL}(P^h \| P^{g+\varepsilon}) = \log \frac{c_g + \varepsilon}{c_h} + \int \log h(x_1) P^h(x_0, dx_1) - \int \log(g(x_1) + \varepsilon) P^h(x_0, dx_1).$$

As  $\varepsilon \downarrow 0$  each term has a finite limit provided  $\int |\log g(x_1)| P^h(x_0, dx_1) < \infty$ . Since  $\log g(x_1)$  is quadratic in  $x_1$  (as  $g$  is Gaussian), this requires  $P^h$  to have finite second moments. Hence

$$\text{KL}(P^h \| P^{g+\varepsilon}) \rightarrow \text{KL}(P^h \| P^g) < \infty \quad \text{as } \varepsilon \downarrow 0,$$

whenever  $P^h$  has finite second moments, which holds for any  $\varepsilon \geq 0$  in our specific example.

**4.3.  $\chi^2$  divergence and finite variance: the threshold.** For any  $\varepsilon > 0$ , the denominator  $g(x_1) + \varepsilon \geq \varepsilon > 0$  uniformly, so

$$m(\varepsilon) \leq \varepsilon^{-1} \int h(x_1)^2 P(x_0, dx_1) < \infty,$$

provided  $h \in L^2(P(x_0, \cdot))$ , which holds for our specific  $h$ . Hence the IS estimator has finite variance for *every*  $\varepsilon > 0$ .

At  $\varepsilon = 0$  the situation is more delicate. From the tail analysis of Section 2, the integrand  $h(x_1)^2/g(x_1)$  behaves like

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right) \quad \text{as } t = x_1 - v \rightarrow +\infty,$$

leading to the following result.

**Proposition 2.** For the observation scheme  $v = X_1 + \log Z^2$ ,  $Z \sim \mathcal{N}(0, 1)$ :

$$\chi^2(P^h \| P^g) < \infty \iff m(0) < \infty \iff \eta < \sigma.$$

Now

$$ESS \approx \frac{N}{1 + \chi^2(P^h \| P^{g+\varepsilon})}.$$

So minimising  $m(\varepsilon)$  at each SMC step is equivalent to maximising ESS.

## 5. NUMERICAL EXPERIMENT: EFFECT OF $\varepsilon$ ON SMC PERFORMANCE

5.1. **Setup.** We consider the hidden Markov model

$$\begin{aligned} X_t | X_{t-1} &\sim \mathcal{N}(\omega + \psi X_{t-1}, \eta^2), \\ V_t | X_t &= X_t + \log Z_t^2, \quad Z_t \sim \mathcal{N}(0, 1), \end{aligned}$$

with parameters  $\omega = 0$ ,  $\psi = 0.9$ ,  $\mu = -1.27$ ,  $\sigma^2 = \pi^2/2$ . We compare two regimes:

- **Regime A** ( $\eta = 1.0 < \sigma \approx 2.22$ ): finite-variance regime, where  $m(0) < \infty$ .
- **Regime B** ( $\eta = 3.0 > \sigma$ ): infinite-variance regime, where  $m(0) = \infty$ .

In both regimes we run SMC with the twisted proposal  $P^{g+\varepsilon}$  for several values of

$$\varepsilon \in \{0, 0.001, 0.01, 0.05, 0.1\}.$$

5.2. **SMC Algorithm.** At each time step  $t$ , the twisted proposal is the mixture

$$P^{g+\varepsilon}(x, \cdot) = \frac{c(x)}{c(x) + \varepsilon} P^g(x, \cdot) + \frac{\varepsilon}{c(x) + \varepsilon} P(x, \cdot),$$

where both  $c(x)$  and the parameters of  $P^g$  are available in closed form (see Section 2). The IS weight for a particle  $x_t$  proposed from  $P^{g+\varepsilon}(x_{t-1}, \cdot)$  is

$$G_\varepsilon(x_{t-1}, x_t) = \frac{h_t(x_t)}{g_t(x_t) + \varepsilon} (c(x_{t-1}) + \varepsilon).$$

To assess performance, we run Algorithm 1 independently  $R = 500$  times on the same simulated dataset ( $T = 50$ ,  $N = 2000$ ) and record both the minimum ESS over time and the maximum normalised weight over time for each run. Define

$$\begin{aligned} ESS_{\min} &= \min_{t=1, \dots, T} ESS_t, \\ Q_{\max} &= \max_{t=1, \dots, T} \max_n W_t^n. \end{aligned}$$

A low  $ESS_{\min}$  indicates that at some point the particle cloud collapsed to a near-degenerate distribution, with a single particle carrying almost all the weight. similarly a large  $Q_{\max}$  value indicates this.

5.3. **Results.** Figure 1 shows the empirical CDF of  $ESS_{\min}$  across  $R = 500$  runs for each value of  $\varepsilon$ , separately for the two regimes. We took  $N = 500$  particles for a dataset with  $T = 50$ .

- *Regime A* ( $\eta < \sigma$ ). One can see an improvement upon taking  $\varepsilon > 0$ , though  $\varepsilon = 0.01$  is clearly too large. One can see that the minimal ESS is about 60 for  $\varepsilon = 0$ . This confirms the theoretical prediction: when  $m(0) < \infty$ , the IS weights have finite variance. While finiteness is guaranteed, it is typically beneficial to take nonzero  $\varepsilon$ . The value  $m(0)$  can still be large, while finite.

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**Algorithm 1** Guided particle filter (GPF) with twisted proposal  $P^{g+\varepsilon}$ , following ?, Algorithm 10.5

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**Require:** observations  $v_{1:T}$ , particle count  $N$ , perturbation  $\varepsilon \geq 0$ , ESS threshold  $\text{ESS}_{\min}$

- 1: For  $n = 1, \dots, N$ : draw  $X_0^n \sim P(x_0)$ , set  $w_0^n \leftarrow G_0(X_0^n)$ ,  $W_0^n \leftarrow w_0^n / \sum_{m=1}^N w_0^m$   
 where  $G_0(x_0) := P_0(dx_0) h_0(x_0) / M_0(x_0)$
  - 2: **for**  $t = 1, \dots, T$  **do**
  - 3:   **if**  $\text{ESS}(W_{t-1}^{1:N}) < \text{ESS}_{\min}$  **then**
  - 4:      $A_{t-1}^{1:N} \leftarrow \text{resample}(W_{t-1}^{1:N})$
  - 5:      $\hat{w}_{t-1} \leftarrow 1$
  - 6:   **else**
  - 7:      $A_{t-1}^n \leftarrow n$
  - 8:      $\hat{w}_{t-1} \leftarrow w_{t-1}^n$
  - 9:   **end if**
  - 10:   For  $n = 1, \dots, N$ : draw  $X_t^n \sim M_t(X_{t-1}^{A_{t-1}^n}, dx_t) = P^{g_t+\varepsilon}(X_{t-1}^{A_{t-1}^n}, dx_t)$   
 by sampling from  $P^{g_t}(X_{t-1}^{A_{t-1}^n}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$  with probability  $\alpha^n = c(X_{t-1}^{A_{t-1}^n}) / (c(X_{t-1}^{A_{t-1}^n}) + \varepsilon)$ , and from  $P(X_{t-1}^{A_{t-1}^n}, \cdot)$  otherwise
  - 11:    $w_t^n \leftarrow \hat{w}_{t-1}^n \cdot G_t(X_{t-1}^{A_{t-1}^n}, X_t^n)$   
 where
 
$$G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon} \cdot (c(x_{t-1}) + \varepsilon)$$
  - 12:    $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
  - 13: **end for**
  - 14: **return**  $\sum_{t=1}^T \log \ell_t^N$ ,  $\{\text{ESS}(W_t^{1:N})\}_{t=1}^T$   
 where, following equation (10.3) of ?,
 
$$\ell_t^N := \begin{cases} \frac{1}{N} \sum_{n=1}^N w_t^n & \text{if resampling occurred at time } t, \\ \left( \sum_{n=1}^N w_t^n \right) / \left( \sum_{n=1}^N w_{t-1}^n \right) & \text{otherwise.} \end{cases}$$
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- *Regime B* ( $\eta > \sigma$ ). The picture changes dramatically. At  $\varepsilon = 0$  the CDF has substantial mass near zero, meaning that in a significant fraction of runs the particle cloud collapses to near-degeneracy at some time step. Taking  $\varepsilon = 0.01$  shifts the CDF markedly rightward, essentially eliminating the collapses. This is the direct numerical manifestation of  $m(0) = \infty$ : the IS weights have infinite variance, leading to occasional but catastrophic weight concentration on a single particle.

**5.4. On the difficulty of choosing  $\varepsilon$  in Regime B.** The results highlight that the choice of  $\varepsilon$  in Regime B is delicate. The optimal  $\varepsilon^*$  exists and can in principle be found by minimising  $m(\varepsilon)$  numerically, as described in Section 3. However,  $m(\varepsilon)$  depends on  $x_0$  and varies across time steps as the particle cloud evolves, so a single fixed  $\varepsilon$  cannot be globally optimal. Furthermore, the performance is sensitive to this choice: the 5th

percentile of  $\text{ESS}_{\min}$  drops sharply on both sides of  $\varepsilon^*$ , leaving a narrow window of well-performing values.

This suggests that a fixed  $\varepsilon$  is fundamentally limited in Regime B, and that an *adaptive* scheme — choosing  $\varepsilon_t$  at each step based on the current particle cloud by minimising an empirical estimate of  $m(\varepsilon)$  — would be more robust. In Regime A, by contrast,  $\varepsilon = 0$  is already satisfactory.

**5.5. Work by Chatterjee & Diaconis.** Suggest to look at  $Q_{\max}$ -statistic. See Figure 2. is maximum of the normalised weights. Note that  $Q_{\max} \in [0, 1]$  and we want it to be small.

## 6. ADAPTIVE CHOICE OF $\varepsilon_t^*$

The following result motivates the adaptive estimator. For a fixed particle  $x_{t-1}$  and observation  $v_t$ , under  $X_t \sim P^{g_t+\varepsilon}(x_{t-1}, \cdot)$ , the second moment of the IS weight equals

$$m(\varepsilon, x_{t-1}) = (c(x_{t-1}) + \varepsilon) \int \frac{h_t(x_1)^2}{g_t(x_1) + \varepsilon} P(x_{t-1}, dx_1).$$

Given the particle cloud  $\{X_{t-1}^n\}$  with normalised weights  $\{W_{t-1}^n\}$ , the population-level second moment is the weighted average

$$\tilde{m}(\varepsilon) = \sum_{n=1}^N W_{t-1}^n \cdot m(\varepsilon, X_{t-1}^n) = \sum_{n=1}^N W_{t-1}^n (c(X_{t-1}^n) + \varepsilon) \int \frac{h_t(x_1)^2}{g_t(x_1) + \varepsilon} P(X_{t-1}^n, dx_1).$$

So we evaluate  $m(\varepsilon)$  by taking the expectation of  $x_{t-1}$  over the current particle approximation. The idea is to choose

$$\varepsilon_t^* = \operatorname{argmin}_{\varepsilon \geq 0} \tilde{m}(\varepsilon).$$

The integral appearing in  $\tilde{m}(\varepsilon)$  can be approximately using numerical quadrature, but in our implementation we approximate it by drawing  $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot)$  once and computing

$$\hat{m}(\varepsilon) = \sum_{n=1}^N W_{t-1}^n \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} \cdot (c(X_{t-1}^n) + \varepsilon).$$

Thus, this is a Monte-Carlo estimator for  $\tilde{m}(\varepsilon)$  based on  $N$  samples. The analytical derivative with respect to  $\varepsilon$  is

$$\frac{d\hat{m}}{d\varepsilon} = \sum_{n=1}^N W_{t-1}^n h_t(\tilde{X}_t^n)^2 \frac{g_t(\tilde{X}_t^n) - c(X_{t-1}^n)}{(g_t(\tilde{X}_t^n) + \varepsilon)^2},$$

which enables gradient-based optimisation. An alternative consists of minimising  $\hat{m}(\varepsilon)$  over a log-spaced grid  $\mathcal{E} = \{\varepsilon_1 < \dots < \varepsilon_K\}$ .

Remarks.

- The auxiliary proposals  $\{\tilde{X}_t^n\}$  are drawn from the untwisted kernel  $P$  and discarded after the grid search. Since they do not enter the weight update, the likelihood estimator  $\sum_t \log \ell_t^N$  remains unbiased for  $\log p(v_{1:T})$  for any data-dependent choice of  $\varepsilon_t^*$ .



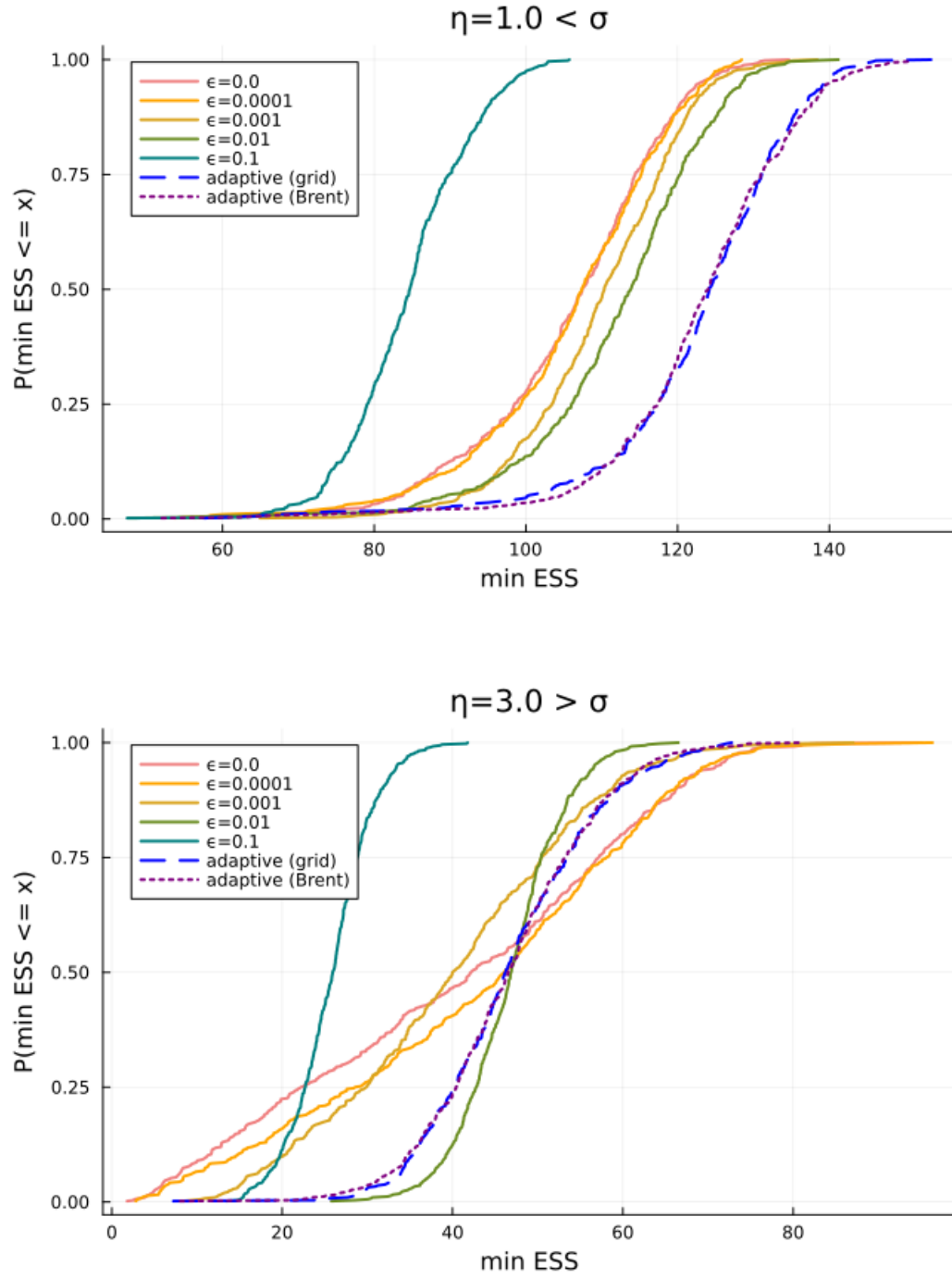


FIGURE 1. minESS comparison.

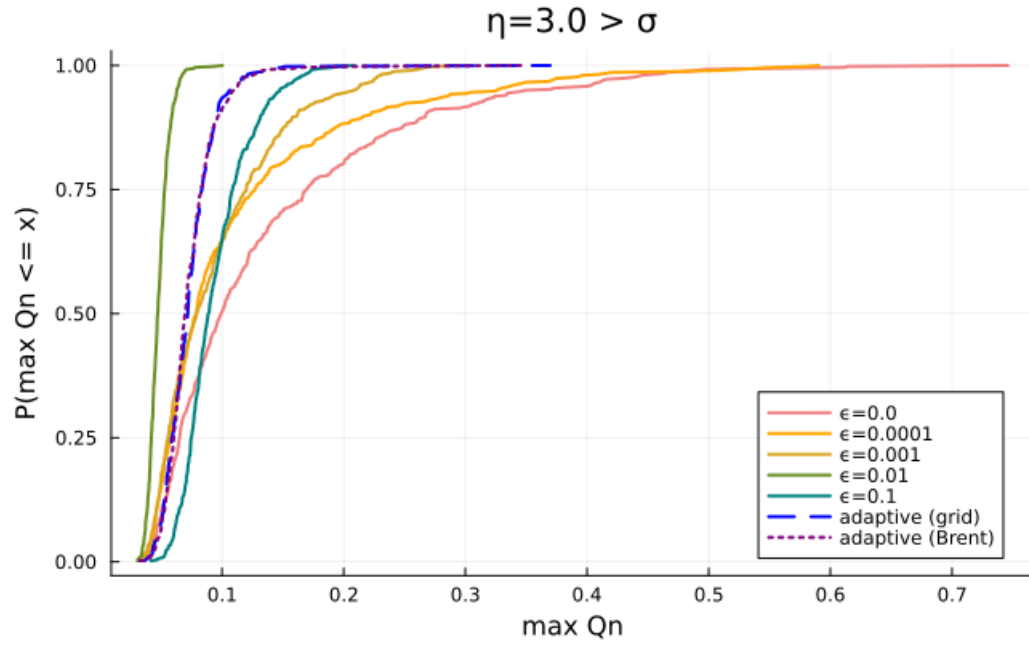
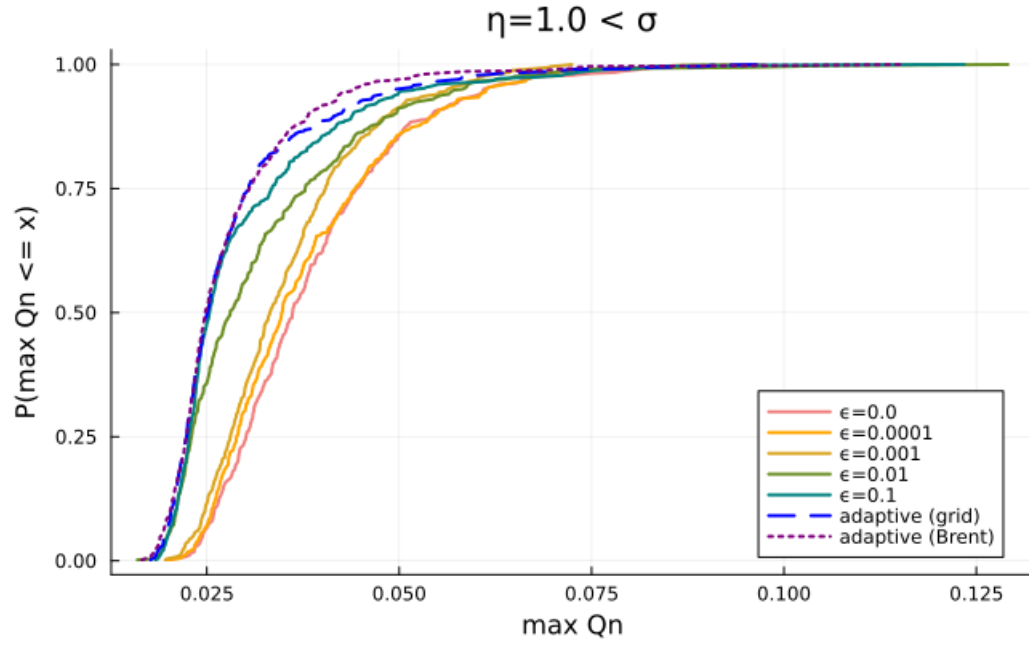


FIGURE 2. Max  $Q_n$  comparison.

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**Algorithm 2** Adaptive GPF with  $\varepsilon_t^*$  chosen by weighted grid search

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**Require:** observations  $v_{1:T}$ , particle count  $N$ , ESS threshold  $\text{ESS}_{\min}$ , grid  $\mathcal{E} = \{\varepsilon_1 < \dots < \varepsilon_K\}$

- 1: For  $n = 1, \dots, N$ : draw  $X_0^n \sim \mathcal{N}(0, \eta^2/(1 - \psi^2))$ , set  $W_0^n \leftarrow 1/N$ ,  $w_0^n \leftarrow 1$
- 2: **for**  $t = 1, \dots, T$  **do**
- 3:   **Find**  $\varepsilon_t^*$ : draw auxiliary proposals  $\tilde{X}_t^n \sim P(X_{t-1}^n, \cdot)$  for  $n = 1, \dots, N$ , and set

$$\varepsilon_t^* = \underset{\varepsilon \in \mathcal{E}}{\operatorname{argmin}} \hat{m}(\varepsilon), \quad \hat{m}(\varepsilon) = \sum_{n=1}^N W_{t-1}^n \frac{h_t(\tilde{X}_t^n)^2}{g_t(\tilde{X}_t^n) + \varepsilon} \cdot (c(X_{t-1}^n) + \varepsilon)$$

Discard  $\{\tilde{X}_t^n\}$  after this step.

- 4:   **if**  $\text{ESS}(W_{t-1}^{1:N}) < \text{ESS}_{\min}$  **then**
- 5:      $A_{t-1}^{1:N} \leftarrow \text{resample}(W_{t-1}^{1:N})$ ,  $\hat{w}_{t-1}^n \leftarrow 1$
- 6:   **else**
- 7:      $A_{t-1}^n \leftarrow n$ ,  $\hat{w}_{t-1}^n \leftarrow w_{t-1}^n$
- 8:   **end if**
- 9:   **Propose:** for  $n = 1, \dots, N$  draw  $X_t^n \sim P^{g_t + \varepsilon_t^*}(X_{t-1}^{A_{t-1}^n}, \cdot)$  by sampling from  $P^{g_t}(X_{t-1}^{A_{t-1}^n}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$  with probability  $\alpha^n = c(X_{t-1}^{A_{t-1}^n})/(c(X_{t-1}^{A_{t-1}^n}) + \varepsilon_t^*)$ , and from  $P(X_{t-1}^{A_{t-1}^n}, \cdot)$  otherwise
- 10:   **Update weights:**

$$w_t^n = \hat{w}_{t-1}^n \cdot G_t(X_{t-1}^{A_{t-1}^n}, X_t^n), \quad G_t(x_{t-1}, x_t) := \frac{h_t(x_t)}{g_t(x_t) + \varepsilon_t^*} \cdot (c(x_{t-1}) + \varepsilon_t^*)$$

- 11:   **Likelihood increment** (eq. 10.3 of ?):

$$\ell_t^N := \begin{cases} \frac{1}{N} \sum_{n=1}^N w_t^n & \text{if resampling occurred at time } t, \\ \left( \sum_{n=1}^N w_t^n \right) / \left( \sum_{n=1}^N w_{t-1}^n \right) & \text{otherwise} \end{cases}$$

- 12:   **Normalise:**  $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$
  - 13:   **ESS:**  $\text{ESS}_t \leftarrow (\sum_n (W_t^n)^2)^{-1}$
  - 14: **end for**
  - 15: **return**  $\sum_{t=1}^T \log \ell_t^N$ ,  $\{\text{ESS}_t\}_{t=1}^T$ ,  $\{\varepsilon_t^*\}_{t=1}^T$
- 

- Using the carry-over weights  $W_{t-1}^n$  in  $\hat{m}(\varepsilon)$  is essential when resampling has not occurred recently. After resampling all weights equal  $1/N$  and the weighted and unweighted averages coincide; in steps without resampling the weight distribution can be skewed and the unweighted average gives a poorly-targeted  $\varepsilon_t^*$ . Numerical experiments confirm that the weighted version gives a substantial improvement in both min-ESS and  $Q_{\max}$  diagnostics.
- The grid  $\mathcal{E}$  is log-spaced over  $[10^{-6}, 1]$  with  $K = 50$  points. All  $K$  evaluations of  $\hat{m}(\varepsilon)$  reuse the same proposals  $\{\tilde{X}_t^n\}$ , so the cost is  $O(KN)$  per step. Reducing to  $K = 20$  or  $K = 30$  would roughly halve the cost with negligible impact on  $\varepsilon_t^*$ .

## 7. ADAPTIVE CHOICE OF $\varepsilon$ VIA ESS TARGETING

This does not work; I guess the desired ESS simply cannot be attained.

**7.1. Motivation.** The numerical experiments of Section 4 show that the optimal perturbation  $\varepsilon^*$  varies across time steps and depends on the current particle cloud. A fixed  $\varepsilon$  cannot be globally optimal. We propose an adaptive scheme inspired by tempering methods in SMC ??, where a tempering parameter is chosen at each step so that the ESS after reweighting equals a target fraction  $\rho N$ . The same principle applies here: we choose  $\varepsilon_t^*$  such that the ESS of the IS weights equals  $\rho N$ .

**7.2. ESS as a Function of  $\varepsilon$ .** Given proposals  $\{x_t^{(i)}\}_{i=1}^N$  drawn from  $P(x_{t-1}^{(i)}, \cdot)$ , the ESS of the weights  $w^{(i)}(\varepsilon) = h_t(x_t^{(i)})/(g_t(x_t^{(i)}) + \varepsilon)$  is

$$\text{ESS}(\varepsilon) = \frac{(\sum_i w^{(i)}(\varepsilon))^2}{\sum_i w^{(i)}(\varepsilon)^2}.$$

Two key properties make this suitable for bisection:

- (1)  $\text{ESS}(\varepsilon)$  is *increasing* in  $\varepsilon$ : as  $\varepsilon \rightarrow \infty$ , all weights converge to  $h_t(x_t^{(i)})/\varepsilon$  and become equal, so  $\text{ESS} \rightarrow N$ .
- (2)  $\text{ESS}(0)$  may be small or even zero if some  $g_t(x_t^{(i)}) = 0$ , reflecting the weight degeneracy in the infinite-variance regime.

The target  $\varepsilon_t^*$  is therefore the unique solution of

$$\text{ESS}(\varepsilon_t^*) = \rho N,$$

found by bisection on  $\varepsilon \in [0, \varepsilon_{\max}]$ . If  $\text{ESS}(0) \geq \rho N$  already, we set  $\varepsilon_t^* = 0$ ; if even  $\varepsilon_{\max}$  is insufficient, we set  $\varepsilon_t^* = \varepsilon_{\max}$ .

**7.3. Connection to  $m(\varepsilon)$ .** The ESS targeting criterion is closely related to minimising  $m(\varepsilon)$ . Note that

$$\frac{\text{ESS}(\varepsilon)}{N} \approx \frac{1}{1 + \chi^2(P^h \| P^{g+\varepsilon})} = \frac{c_h(x_0)^2}{m(\varepsilon)},$$

so targeting  $\text{ESS}(\varepsilon) = \rho N$  is equivalent to targeting  $\chi^2(P^h \| P^{g+\varepsilon}) = (1 - \rho)/\rho$ , i.e. a fixed level of the  $\chi^2$  divergence. Rather than minimising  $m(\varepsilon)$  (which requires a grid search and gives a different target at each step), ESS targeting gives a directly interpretable, step-by-step criterion for particle diversity, and bisection finds the solution efficiently without a grid.

## 8. CONCLUSION

Figure 3 shows the results of a single SMC run for both regimes and both fixed ( $\varepsilon = 0$ ) and adaptive  $\varepsilon^*$ . Even for  $\eta = 3$  the results with  $\varepsilon = 0$  can be just fine: a single run looks almost indistinguishable from the adaptive run in terms of ESS, cumulative log-likelihood, and incremental log-likelihood. However, if one were to repeat the analysis many times, occasionally this would result in a catastrophic drop in ESS. This is precisely the implication of  $m(0) = \infty$ : it does not mean every run fails, but that the variance of the IS weights across runs is infinite, so disasters occur with positive probability and cannot be ruled out from a single run alone. This is confirmed by the CDF plots of

Figure 1, which required  $R = 200$  independent runs to reveal the heavy left tail of the min-ESS distribution at  $\varepsilon = 0$ .

Two further observations support this interpretation. First, the incremental log-likelihood panel shows a pronounced dip around  $t = 30$  in both the fixed and adaptive runs for  $\eta = 3$ . This dip is driven by a particular difficult observation and occurs regardless of  $\varepsilon$ , suggesting it reflects a genuinely hard step in the data rather than an algorithmic failure. In a catastrophic run, this dip would be far deeper and the ESS would collapse to near 1 at that point. Second, the adaptive  $\varepsilon^*$  panel for  $\eta = 3$  shows  $\varepsilon^*$  spiking upward precisely around  $t = 30$ , at the same difficult observation. This confirms that the adaptive scheme is responding correctly: it detects the hard step through the empirical minimiser of  $\hat{m}(\varepsilon)$  and automatically increases  $\varepsilon$  to protect against weight collapse, without any explicit knowledge of which observations are difficult.

## 9. LAPLACE OBSERVATION SCHEME

**9.1. Setup.** As an alternative to the log-chi-squared observation scheme, consider

$$V_t = X_t + U_t, \quad U_t \sim \text{Laplace}(0, b),$$

so that the observation density evaluated at  $v$  is

$$h(x_1) = \frac{1}{2b} \exp\left(-\frac{|v - x_1|}{b}\right).$$

We retain the Gaussian approximation

$$g(x_1) = \varphi(x_1; v - \mu, \sigma^2),$$

so all closed-form expressions for  $c(x_0)$ ,  $P^g$ , and the mixture sampler carry over unchanged from the Gaussian observation case.

**9.2. Tail analysis of  $m(0)$ .** Writing  $t = x_1 - v$ , the integrand of  $I(0) = \int h(x_1)^2 / g(x_1) P(x_0, dx_1)$  behaves as

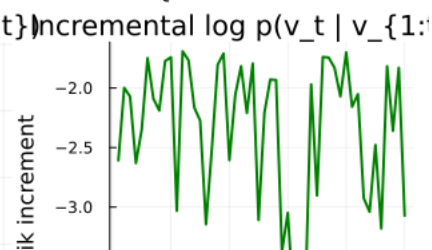
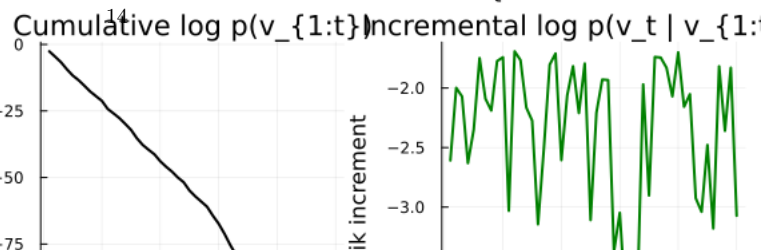
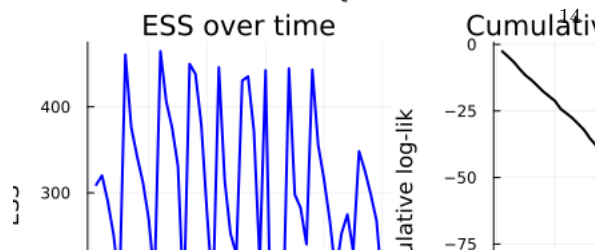
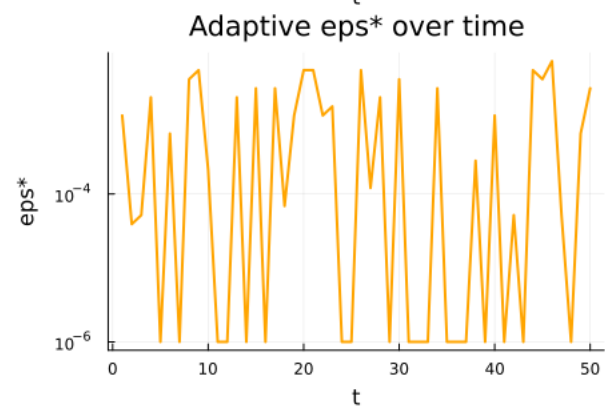
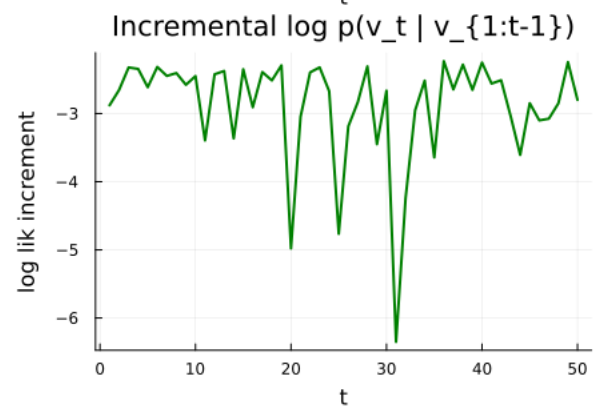
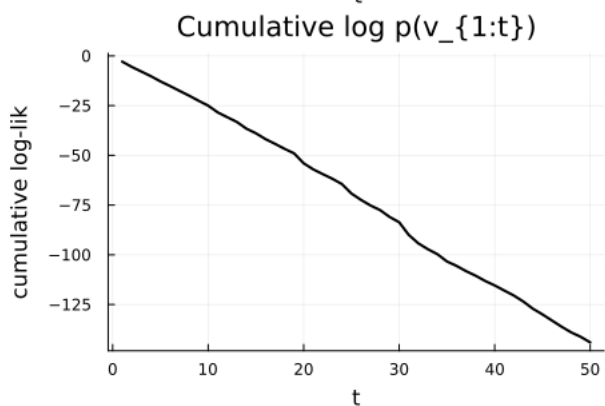
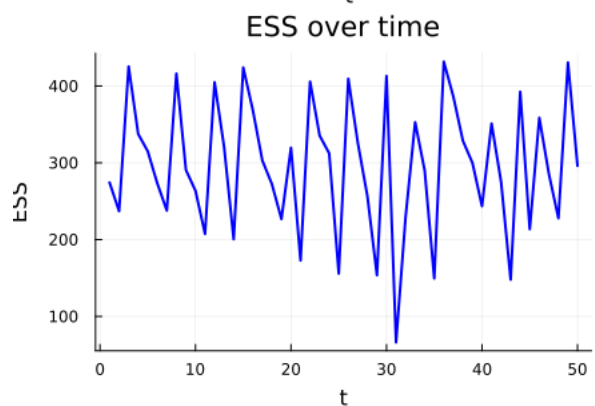
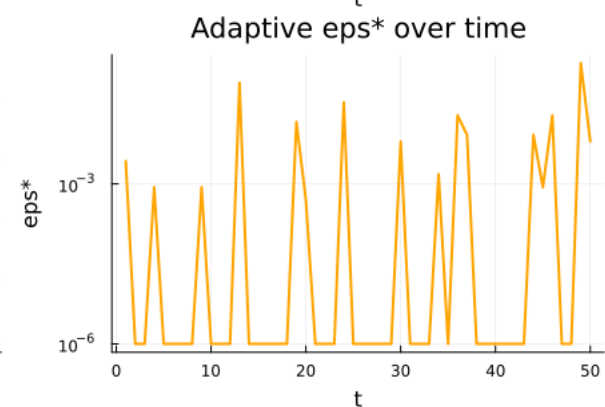
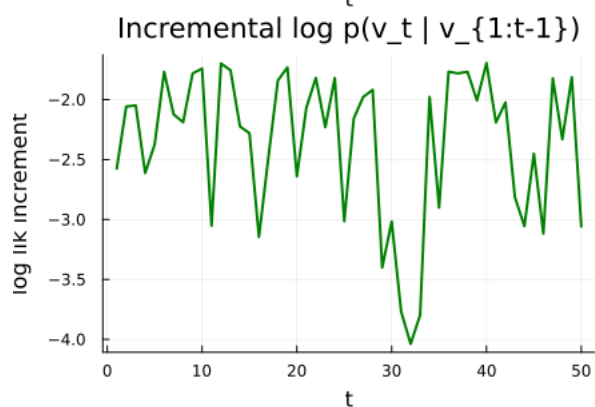
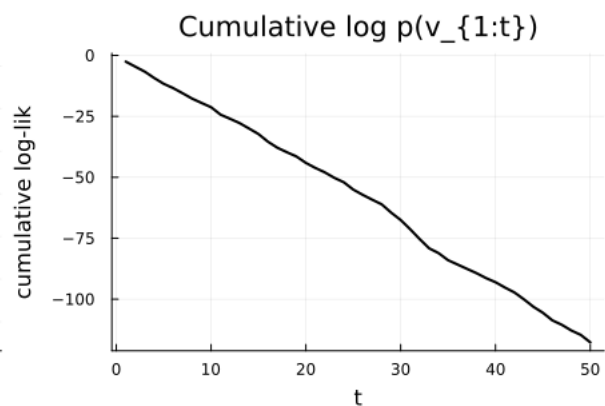
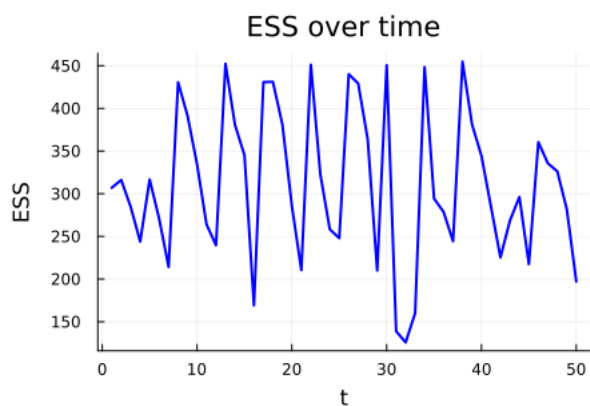
$$\frac{h(x_1)^2}{g(x_1)} p(x_1 | x_0) \propto \exp\left(-\frac{2|t|}{b} + \frac{(t + \mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2}\right).$$

- *Right tail*  $t \rightarrow +\infty$ . The linear term  $-2t/b$  and the quadratic terms combine to give dominant behaviour

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) - \frac{2t}{b} + O(t)\right).$$

For this to be integrable we need the coefficient of  $t^2$  to be strictly negative, i.e.  $\eta^2 < \sigma^2$ , exactly as in the Gaussian observation case. When  $\eta^2 \geq \sigma^2$ , the quadratic growth dominates the linear decay from the Laplace tails and  $I(0) = \infty$ .

When  $\eta^2 < \sigma^2$  the quadratic term decays, but the linear term  $-2t/b$  provides additional decay that is absent in the Gaussian observation case. The integrand therefore decays faster here, but  $I(0)$  is still finite only when  $\eta^2 < \sigma^2$ .



- *Left tail*  $t \rightarrow -\infty$ . The Laplace density contributes  $-2|t|/b = +2t/b \rightarrow -\infty$ , providing exponential decay. However, when  $\eta^2 \geq \sigma^2$  the quadratic term grows, and whether it dominates depends on the sign of  $1/(2\sigma^2) - 1/(2\eta^2)$ . When  $\eta^2 \geq \sigma^2$  the quadratic growth again wins and  $I(0) = \infty$ .

**Proposition 3.** For the Laplace observation scheme  $V_t = X_t + U_t$ ,  $U_t \sim \text{Laplace}(0, b)$ :

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1) < \infty \iff \eta^2 < \sigma^2.$$

Consequently,  $\bar{m}(0)$  is finite if and only if  $\eta < \sigma$ , independently of the scale parameter  $b$ .

The threshold  $\eta = \sigma$  is the same as for the Gaussian observation scheme. However, the Laplace tails make  $I(0)$  larger for any fixed  $\eta < \sigma$  compared to the Gaussian case, meaning the need for  $\varepsilon > 0$  is more pressing even in the finite-variance regime. Furthermore, larger  $\eta$  increases the coefficient of  $t^2$ , making  $I(\varepsilon)$  larger for any fixed  $\varepsilon > 0$  and the optimal  $\varepsilon^*$  correspondingly larger.

**9.3. The limit  $b \downarrow 0$ .** As the scale  $b \rightarrow 0$ , the Laplace distribution concentrates on zero and the observation scheme approaches a noiseless observation  $V_t = X_t$ . In this limit:

- $h(x_1) = \frac{1}{2b} \exp(-|v - x_1|/b) \rightarrow \delta_{x_1=v}$ , a point mass at  $x_1 = v$ .
- The twisted target  $P^h(x_0, \cdot)$  concentrates all mass at  $x_1 = v$ , i.e. the posterior on  $X_t$  given  $V_t = v$  becomes a point mass.
- The ratio  $h(x_1)/g(x_1)$  becomes extremely large near  $x_1 = v$  and extremely small away from it, since  $h$  is an increasingly sharp spike while  $g$  remains spread out. This makes the IS weights degenerate: a single particle near  $v$  gets essentially all the weight.
- Consequently,  $m(\varepsilon) \rightarrow \infty$  for any fixed  $\varepsilon \geq 0$  as  $b \rightarrow 0$ , and the optimal  $\varepsilon_b^* \rightarrow \infty$  as well — no finite regularisation can rescue the IS estimator when the observation is nearly noiseless.

This limiting behaviour highlights a fundamental tension: the more informative the observation (smaller  $b$ ), the harder the IS problem becomes. Intuitively, the Gaussian approximation  $g$  is a poor proxy for a nearly point-mass likelihood, and  $\varepsilon$  can only partially compensate by mixing in the prior  $P$ . In the limit  $b = 0$ , importance sampling from  $P^{g+\varepsilon}$  breaks down entirely and a different approach — such as an exact update using the point-mass likelihood — would be needed.

Numerically, one can verify this by plotting  $\bar{m}(\varepsilon)$  for decreasing values of  $b$ : the minimum of  $\bar{m}$  increases without bound and  $\varepsilon^*$  drifts toward larger values as  $b \rightarrow 0$ .

## 10. PUSHING THE ARGUMENT TO SDEs

Suppose

$$dX_t = b(t, X_t) dt + q dW_t, \quad X_0 = x_0. \quad (2)$$

Here,  $b: \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a (typically nonlinear) map known as the drift function and  $q \in \mathbb{R}^{d \times d'}$  is the diffusion function. Furthermore,  $W$  is a standard Brownian motion process in  $\mathbb{R}^{d'}$ . Assume we observe  $v$  at time  $T$  and let

$$h(x_T) = \text{density of } v \text{ conditional on } X_T = x_T.$$

We can

- Make a Gaussian approximation to  $h(x_T)$ .
- Compute  $g(t, x)$  for all  $t$ .
- Use guided proposals with not  $g$  but  $g + \varepsilon$ .

Rather than  $g$ , take  $g_\epsilon := g + \epsilon$ . Then

$$\begin{aligned} r_\epsilon(t, x) &:= \nabla_x g_\epsilon(t, x) = \frac{g'(t, x)}{g(t, x) + \epsilon} = \frac{g(t, x)}{g(t, x) + \epsilon} \nabla_x \log g(t, x) \\ &= s(-\log \epsilon + \log g(t, x)) \nabla_x \log g(t, x), \end{aligned}$$

where  $s(x) = 1/(1 + e^{-x})$ . Thus we get a damping factor. The measure  $P^{g+\epsilon}$  is the measure (on path space) of the process

$$dX_t = b(t, X_t) dt + Q r_\epsilon(t, X_t) + q dW_t, \quad X_0 = x_0.$$

with  $Q = qq^\top$ . Now, with  $x = (x_t, t \in [0, T])$ .

$$G_\epsilon(x) = h(x_T) \frac{dP}{dP^{g+\epsilon}}(x).$$

Here

$$\frac{dP}{dP^{g+\epsilon}}(x) = \frac{g(x_0) + \varepsilon}{g(x_T) + \varepsilon} \Psi(x),$$

with

$$\log \Psi(x) = \int_0^T (b(t, x_t) - \tilde{b}(t, x_t)) r_\epsilon(t, x_t) dt.$$

Thus

$$\begin{aligned} m(\varepsilon) &= \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X)^2 = \int G_\varepsilon(x)^2 dP^{g+\varepsilon}(x) \\ &= \int G_\varepsilon(x)^2 \frac{dP^{g+\varepsilon}}{dP}(x) dP(x) \\ &= \int \left( \frac{dP^{g+\varepsilon}}{dP}(x) \right)^{-1} h(x_T)^2 dP(x) \\ &= (g(x_0) + \varepsilon) \int \frac{h(x_T)^2}{g(x_T) + \varepsilon} \Psi(x)^{-1} dP(x). \end{aligned}$$

This is to be compared to (1):  $c_g$  in that formula is exactly as  $g(x_0)$  here. In the discrete-time we assumed linear dynamics and thus  $\tilde{b} = b$  which amounts to  $\Psi(x) = 1$ . **I think the nonlinear case is most interesting here, for else filtering at time  $T$  boils down to the discrete-time problem.**

**Lemma 4.** Suppose  $g(T, x) = \varphi(v; \mu, \sigma^2)$  (with  $\mu, \sigma$  depending on  $v$ ). If  $b(x) = \omega + \psi x$  then for  $t \in [0, T)$  we have

$$g(t, x) = \dots$$

**Closed form expression or via backward ODEs.**

Can we establish when  $m(0) = \infty$ ?